# EIE-IST-Star Q1

- Is this valid code? If it is invalid, explain what is the problem.

```
public class Square_Q1a {
  public int square(int x) { return x*x; }
  public double square(int y) { return y*y; }
}
```

```
public class Square_Q1b {
  public double square(int x) { return x*x; }
  public double square(double y) { return y*y; }
}
```

```
public class Square_Q1c {
  public double square(int x) { return x*x; }
  public int square(double y) { return y*y; }
}
```

# EIE-IST-Star Q2

- a) What is the output of this program? Explain your answer.

```
public class Test_Q2 {
  public static void main(String[] args) {
    String s = "6";
    int n = 3;
    double d = 4.5;
    System.out.println(s + n + d);
  }
}
```

b) Add a pair of brackets in the `println` statement to achieve a different output to a). Explain your answer.

# EIE-IST-Star Q3

- Identify which statements are TRUE and which are FALSE. Justify your answers.
  - ❑ A subclass has direct access to its superclass' private data and methods.
  - ❑ A class can extend more than one superclass.
  - ❑ An abstract class must contain at least one abstract method.
  - ❑ An abstract class must not contain any instance variables.